

第十一章复习题

$$1. \begin{cases} x = 2t + \ln t^2 \\ y = (t + \ln t)^2 \end{cases}, t > 0 \Rightarrow \frac{dy}{dx} = \frac{dy/dt}{dx/dt} = \frac{2(t + \ln t) \cdot (1 + \frac{1}{t})}{2 + 2 \ln t \cdot \frac{1}{t}} = \frac{t+1}{t}$$

$$\frac{d^2y}{dx^2} = \frac{dy'/dt}{dx/dt} = \frac{1}{2 + 2 \ln t \cdot \frac{1}{t}} = \frac{t}{2(t + \ln t)}$$

$$\Rightarrow \frac{d^2y}{dx^2} \Big|_{t=1} = \frac{1}{2}$$

$$2. \begin{cases} x = \cos^3 t \\ y = \sin^3 t \end{cases}, 0 \leq t \leq \frac{\pi}{2} \Rightarrow \frac{dx}{dt} = -3\cos^2 t \cdot \sin t, \frac{dy}{dt} = 3\sin^2 t \cos t$$

$$(1) L = \int_0^{\frac{\pi}{2}} \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2} dt = \int_0^{\frac{\pi}{2}} \sqrt{9\cos^4 t \sin^2 t + 9\sin^4 t \cos^2 t} dt = \int_0^{\frac{\pi}{2}} 3\cos t \sin t dt$$

$$= \frac{3}{2} \int_0^{\frac{\pi}{2}} \sin 2t dt = -\frac{3}{4} \cos 2t \Big|_0^{\frac{\pi}{2}} = \frac{3}{2}$$

$$(2) S = \int_0^{\frac{\pi}{2}} 2\pi y \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2} dt = \int_0^{\frac{\pi}{2}} 2\pi \sin^3 t \cdot 3\sin t \cos t dt = \int_0^{\frac{\pi}{2}} 6\pi \sin^4 t \cos t dt$$

$$= \frac{6}{5} \pi \sin^5 t \Big|_0^{\frac{\pi}{2}} = \frac{6}{5} \pi$$

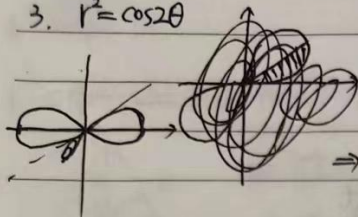
$$3. L = 4 \int_0^{\frac{\pi}{2}} \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2} dt = 6 \text{ (同上2(1))}$$

考友 = :

$$1. r = \csc \theta \cdot e^{\cos \theta} \Rightarrow \frac{1}{\sin \theta} + \sin \theta = e^{\cos \theta} \Rightarrow y = e^x$$

$$2. r = 2\sin \theta \Rightarrow r^2 = 2r\sin \theta \Rightarrow x^2 + y^2 = 2y \Rightarrow x^2 + (y-1)^2 = 1 \quad \checkmark$$

3. $r^2 = \cos 2\theta$

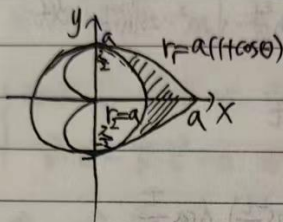


$2\theta = 0 \Rightarrow \theta = 0, r = \pm 1$

$2\theta = \frac{\pi}{2} \Rightarrow \theta = \frac{\pi}{4}, r = 0$

$\Rightarrow A = 4 \int_0^{\frac{\pi}{4}} \frac{1}{2} r^2 d\theta = 4 \int_0^{\frac{\pi}{4}} \frac{1}{2} \cos 2\theta d\theta = \sin 2\theta \Big|_0^{\frac{\pi}{4}} = 1$

$\begin{cases} r_1 = a(1 + \cos \theta) \\ r_2 = a \end{cases}$



(1) $A = \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{1}{2} (r_1^2 - r_2^2) d\theta = \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{1}{2} [a^2(1 + \cos \theta)^2 - a^2] d\theta = \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{1}{2} a^2 (2\cos \theta + \cos^2 \theta) d\theta$

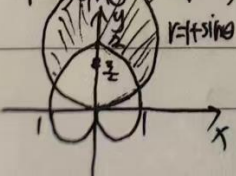
$= \int_0^{\frac{\pi}{2}} a^2 (2\cos \theta + \frac{1 + \cos 2\theta}{2}) d\theta = a^2 (2\sin \theta + \frac{1}{2}\theta + \frac{1}{4}\sin 2\theta) \Big|_0^{\frac{\pi}{2}} = (2 + \frac{\pi}{4})a^2$

(2) $A_1 = 2 \int_0^{\pi} \frac{1}{2} r_1^2 d\theta = \int_0^{\pi} a^2 (1 + \cos \theta)^2 d\theta = \int_0^{\pi} a^2 (1 + 2\cos \theta + \cos^2 \theta) d\theta$

$= \int_0^{\pi} (1 + 2\cos \theta + \frac{1 + \cos 2\theta}{2}) d\theta = \left[\frac{3}{2}\theta + 2\sin \theta \right]_0^{\pi} = \frac{3\pi}{2}a^2$

$\Rightarrow A = A_1 - (2 + \frac{\pi}{4})a^2 = \frac{3\pi}{2}a^2 - (2 + \frac{\pi}{4})a^2 = (\frac{5\pi}{4} - 2)a^2$

5. $r = 3\sin \theta \Rightarrow r^2 = 3r\sin \theta \Rightarrow x^2 + y^2 = 3y \Rightarrow x^2 + (y - \frac{3}{2})^2 = \frac{9}{4}$



$\begin{cases} r = 3\sin \theta \\ r = 1 + \sin \theta \end{cases} \Rightarrow 1 + \sin \theta = 3\sin \theta \Rightarrow \sin \theta = \frac{1}{2} \Rightarrow \theta = \frac{\pi}{6}$
 $A = 2 \int_{\frac{\pi}{6}}^{\frac{\pi}{2}} \frac{1}{2} [(1 + \sin \theta)^2 - (3\sin \theta)^2] d\theta$
 $= \int_{\frac{\pi}{6}}^{\frac{\pi}{2}} (1 + 2\sin \theta - 8\sin^2 \theta) d\theta$